1.4 Time Measurements

Fundamental to both attitude and orbit calculations is the measurement of time and time intervals. Unfortunately, a variety of time systems are in use and sorting them out can cause considerable confusion. A technical discussion of time systems needed for precise computational work is given in Appendix J. In this section, we summarize aspects that are essential to the interpretation of attitude data.

Two basic types of time measurements are used in attitude work: (1) *time intervals* between two events, such as the spacecraft spin period or the length of time a sensor sees the Earth; and (2) *absolute times*, or *calendar times*, of specific events, such as the time associated with some particular spacecraft sensing. Of course, calendar time is simply a time interval for which the beginning event is an agreed standard.

Calendar time in the usual form of date and time is used only for input and output, since arithmetic is cumbersome in months, days, hours, minutes, and seconds. Nonetheless, this is used for most human interaction with attitude systems because it is the system with which we are most familiar. Problems arise even with date and time systems, since time zones are different throughout the world and spacecraft operations involve a worldwide network. The uniformly adopted solution to this problem is to use the local time corresponding to 0 deg longitude as the standard absolute time for events anywhere in the world or in space. This is referred to as *Universal Time (UT)*, *Greenwich Mean Time (GMT)*, or *Zulu (Z)*, all of which are equivalent for the practical purposes of data interpretation. The name

Greenwich Mean Time is used because 0 deg longitude is defined as going through the site of the former Royal Greenwich Observatory in metropolitan London. Eastern Standard Time in the United States is obtained by subtracting 5 hours from the Universal Time.

Because calendar time is inconvenient for computations, we would like an absolute time that is a continuous count of time units from some arbitrary reference. The time interval between any two events may then be found by simply subtracting the absolute time of the first event from that of the second event. The universally adopted solution for astronomical problems is the *Julian day*, a continuous count of the number of days since noon (12:00 UT) on January 1, 4713 BC. This strange starting point was suggested by an Italian scholar of Greek and Hebrew, Joseph Scaliger, in 1582, as the beginning of the current *Julian period* of 7980 years. This period is the product of three numbers: the solar cycle, or the interval at which all dates recur on the same days of the week (28 years); the lunar cycle containing an integral number of lunar months (19 years); and the indiction, or the tax period introduced by the emperor Constantine in 313 AD (15 years). The last time that these started together was 4713 BC and the next time will be 3267 AD. Scaliger was interested in reducing the astronomical dating problems associated with calendar reforms of his time and his proposal had the convenient selling point that it predated the ecclesiastically approved date of creation, October 4, 4004 BC. The Julian day was named after Scaliger's father, Julius Caeser Scaliger, and was not associated with the Julian calendar that had been in use for some centuries.

Tabulations of the current Julian date may be found in nearly any astronomical ephemeris or almanac. A particularly clever procedure for finding the Julian date, or JD, associated with any current year (I), month (J), and the day of the month (K), is given by Fliegel and Van Flandern [1968] as a FORTRAN arithmetic statement using FORTRAN integer arithmetic:

$$\boxed{\begin{aligned} JD = K - 32075 + 1461 * (I + 4800 + (J - 14)/12)/4 \\ + 367 * (J - 2 - (J - 14)/12 * 12)/12 \\ - 3 * ((I + 4900 + (J - 14)/12)/100)/4 \end{aligned}}$$

Uso de aritmé-
tica inteira do
FORTRAN.
Como é isso?
1) respeite a
hierarquia:
operações de
multiplicação e
divisão tem
prioridade "da
esquerda
p/ a direita";
2) o valor da
operação é o
menor
valor inteiro
associado ao
resultado, na
direção do zero.
Isso equivale a
tomar a divisão
interia e pegar só a
parte inteira,
desprezando a parte
decimal. O comando
'fix' do MATLAB, faz
isso com uma
divisão. Teste-o e
veja.

For example, December 25, 1981 ($I = 1981$, $J = 12$, $K = 25$) is JD 2,444,964.*

The Julian date presents minor problems for space applications. It begins at noon Universal Time, rather than 0 hours Universal Time and the extra digits required for the early starting date limit the precision of times in computer storage. However, no generally accepted substitute exists, and the Julian day remains the only unambiguous continuous time measurement for general use.

For internal computer calculations, the problem of ambiguity does not arise and several systems are used. The *Julian Day for Space*, or *JDS*, is defined as JD − 2,436,099.5. This system starts at 0 hours UT (rather than noon), September 17, 1957, which is the first Julian day divisible by 100 prior to the launch of the first

* Note that in FORTRAN integer arithmetic, multiplication and division are performed left to right, the magnitude of the result is truncated to the lower integer value after each operation, and −2/12 is truncated to 0.

Por ex, -2/12 = -0,1666... => 0
É o menor inteiro associado ao resultado.
Equivale a tomar só a parte inteira do resultado.
o comando 'fix' do MATLAB faz isso.

Importante: definido p/ uso por astrônomos (durante a noite). Quer dizer, considera a noite do dia calendário considerado, para o qual se deseja calcular o dia juliano. Por exemplo, que dia juliano é hoje (após as 18h)? Veja que a resposta é um número inteiro...

manmade satellite by the Soviet Union on October 4, 1957. This system is used internally in NASA orbit programs, with time measured in seconds rather than days. (Measuring all times and time intervals in seconds is convenient for computer use because the large numbers involved do not pose a problem.) The European Space Operations Center uses the *Modified Julian Day*, which starts at 0 hours UT, January 1, 1950. Attitude determination programs at NASA's Goddard Space Flight Center measure time intervals in seconds from 0 hours UT, September 1, 1957. Unfortunately, the origin of this particular system appears to have been lost in antiquity.

O Brasil também usa a DJM.

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